SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 7.2

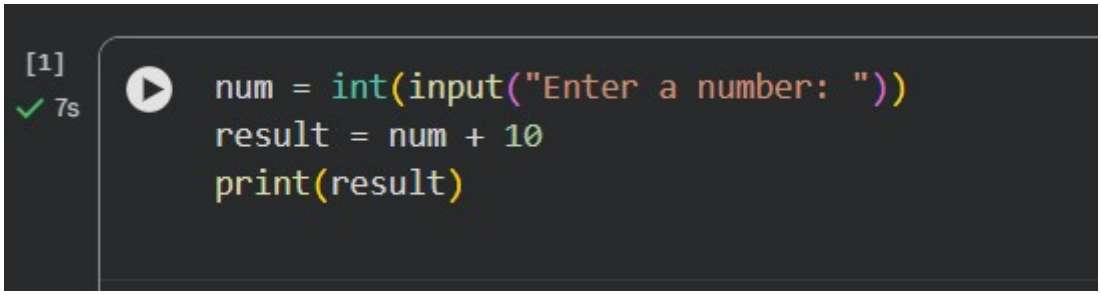
Program	**: B. Tech (CSE)**
Specialization	**:**
Course Title	**: AI Assisted coding**
Course Code	**:**
Semester	**: II**
Academic Session	**: 2025-2026**
Name of Student	**:B.Gnaneshwar**
Enrollment No.	**: 2403A51L15**
Batch No.	**: 51**
Date	**:30-01-2026**

TASK 1: Runtime Error Due to Invalid Input Type

Prompt:

Use AI tools to identify the runtime error caused by invalid input type and correct the program so that it executes properly.

Code :



Explanation:

· `input()` takes user input as a **string** by default.

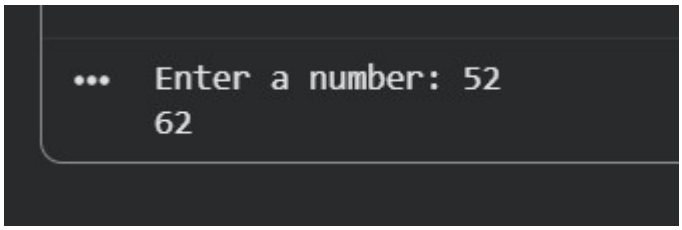
· The program tries to add an integer (`10`) to a string (`num`).

· Python does not allow addition between a string and an integer.

· AI identifies the issue and suggests converting the input to an integer using `int()`.

· After conversion, arithmetic operation works correctly.

Output :

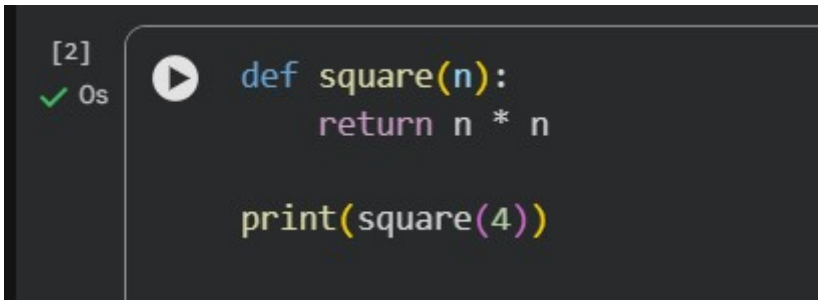


TASK 2: Incorrect Function Return Value

Prompt:

Use AI assistance to analyze the function and ensure that it correctly returns the computed value.

Code :



Explanation:

· The function calculates the square of a number.

· The calculated result is stored in a variable but not returned.

· Without a `return` statement, the function returns `None`.

· AI detects the missing return statement.

· Adding `return` allows the function to send the result back correctly.

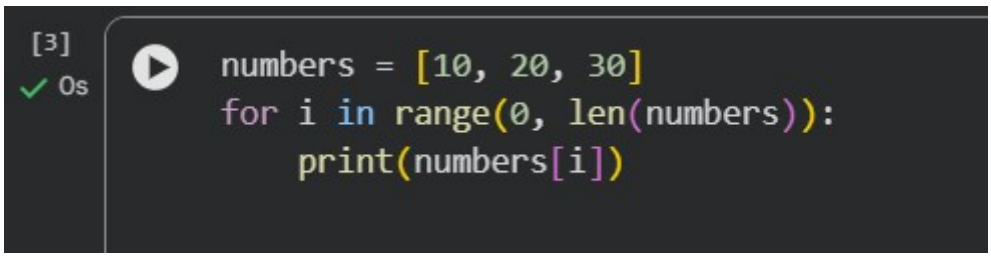
Output :

TASK 3: IndexError in List Traversal

Prompt:

Use AI to identify the incorrect loop boundary and correct the iteration logic.

Code :



Explanation:

· The list contains three elements.

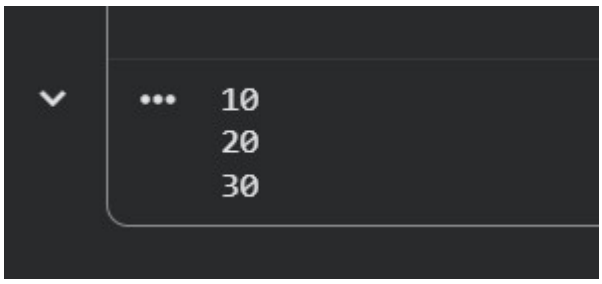
· `len(numbers)` gives the total count of elements.

· Using `len(numbers) + 1` causes the loop to access an invalid index.

· This results in an `IndexError`.

· AI suggests correcting the loop boundary to prevent out-of-range access.

Output :

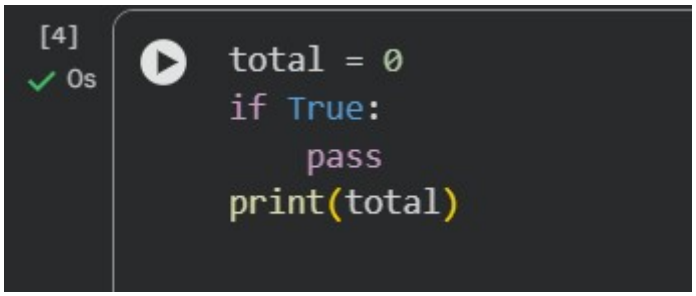


TASK 4: Uninitialized Variable Usage

Prompt:

Use AI tools to detect the uninitialized variable and correct the program.

Code :



Explanation:

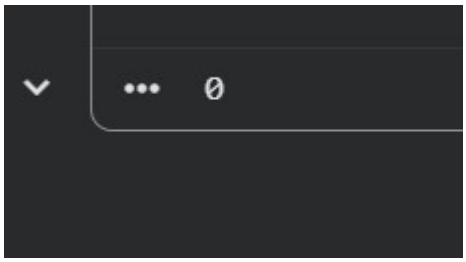
· The variable `total` is used before assigning any value.

· Python raises a `NameError` when an undefined variable is accessed.

· AI identifies the missing initialization.

· Initializing the variable before use fixes the error.

Output :

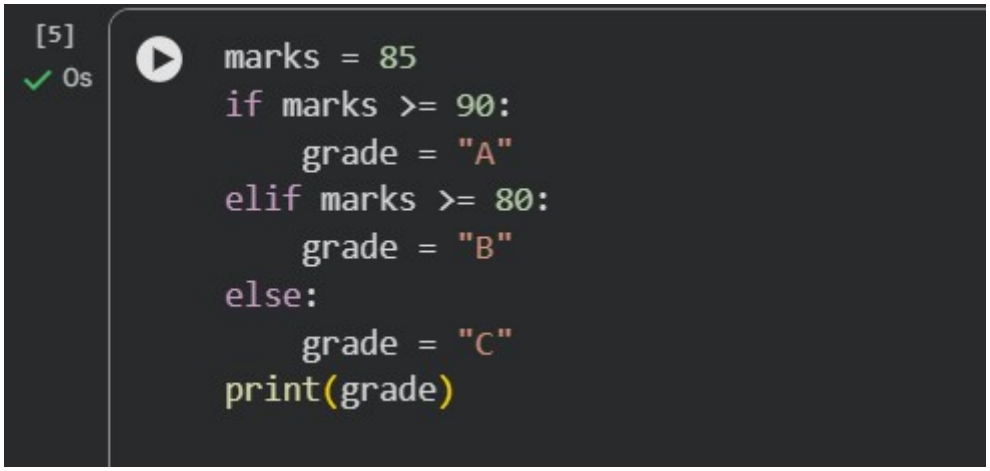


TASK 5: Logical Error in Student Grading System

Prompt:

Use AI to analyze the grading conditions and correct the logical flow.

Code :



Explanation:

· The program assigns grades based on marks.

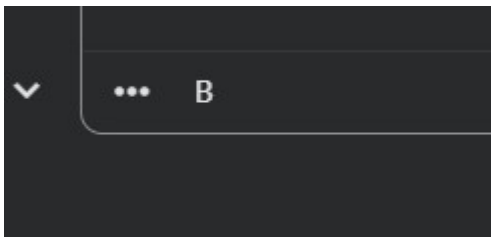
· Marks between 80 and 89 are incorrectly assigned grade "C".

· Logical ordering of conditions is incorrect.

· AI identifies the logical error and rearranges the grading conditions.

· Correct logic assigns grades accurately.

Output :



Final Conclusion (5 Points)

· AI tools help in quickly identifying **syntax, runtime, and logical errors** in Python programs.

· AI provides **clear explanations** for errors, making debugging easier for beginners.

· Using AI improves **systematic debugging skills** and reduces trial-and-error coding.

· AI-assisted suggestions help in **writing cleaner and more reliable code**.

· This lab increased confidence in using AI as an effective **programming support tool**.